

Coding & Solution

Date	1 st July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Solution Summary

Field	Details
Repository Link / URL	https://github.com/SumanthReddy078/Rising-Waters
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Bootstrap 5, Scikit-learn
Key Features Implemented	• Historical weather data preprocessing • Exploratory Data Analysis (EDA) • Feature Scaling using StandardScaler • Random Forest Classification Model • Flask-based Web Application • Responsive User Interface • Flood Risk Prediction • Model Integration using Joblib
Pending / Incomplete Features	• IBM Cloud Deployment (Future Scope) • Real-time Weather API Integration • SMS / Email Alert Notification System
Setup / Run Instructions	1. Install Python dependencies using pip install -r requirements.txt. 2. Run the Flask application using python app.py. 3. Open http://127.0.0.1:5000/ in a web browser. 4. Enter weather parameters and click Predict to obtain the flood prediction result.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions/classes.	Yes
2	Meaningful variable and function names are used.	Yes
3	Code includes comments/documentation where necessary.	Yes
4	Error handling is implemented for critical operations.	Yes
5	The application runs without critical errors.	Yes
6	Code is committed to a version control repository (GitHub).	Yes

Additional Notes / Comments

- The project follows a modular software architecture that separates frontend, backend, machine learning, and data processing components.
- The Random Forest model is trained using historical weather and rainfall data to improve prediction accuracy.
- StandardScaler is applied to preprocess input features before prediction.
- The trained machine learning model and scaler are stored using Joblib for efficient loading during runtime.
- The web application is lightweight, responsive, and designed for future deployment on IBM Cloud.
- The codebase follows software engineering best practices to ensure readability, maintainability, and reusability.